

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date: 8 March 2026

Team ID: (Write your team ID)

Project Name: Quotes Recommendation Chatbot using NLP

Maximum Marks: 4 Marks

Functional Requirements

Following are the **functional requirements of the proposed solution**.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Interaction	User enters text query to the chatbot
FR-2	Input Processing	System receives user input and preprocesses the text
FR-3	NLP Processing	Tokenization and keyword extraction from user query
FR-4	Intent Detection	System identifies the type of quote requested (motivation, love, success, life, funny)
FR-5	Quote Recommendation	System selects a suitable quote from the dataset
FR-6	Response Generation	Chatbot generates a response containing the recommended quote
FR-7	Display Output	Recommended quote is displayed to the user
FR-8	Dataset Management	System stores and retrieves quotes from the dataset

Non-Functional Requirements

Following are the **non-functional requirements of the proposed solution**.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The chatbot interface should be simple and easy for users to interact with.
NFR-2	Security	The system should ensure secure handling of user input and data.
NFR-3	Reliability	The chatbot should provide accurate responses consistently.
NFR-4	Performance	The system should respond to user queries quickly with minimal delay.

NFR No.	Non-Functional Requirement	Description
NFR-5	Availability	The chatbot should be accessible whenever users want to interact with it.
NFR-6	Scalability	The system should support expansion with more quotes and additional features in the future.